

Exp No: 27

BINARY SEMAPHORE IN SYSTEMC

```
GNU nano 8.7.1 binary_semaphore.cpp
#include <systemc.h>

SC_MODULE(binary_semaphore) {
    int sem;
    sc_event ev;

    void task1() {
        while(true) {
            wait(1, SC_SEC);

            if(sem == 1) {
                sem = 0;

                cout << "Task1 Entered Critical Section at "
                     << sc_time_stamp() << endl;

                wait(2, SC_SEC);

                sem = 1;
                ev.notify();
            }
            else {
                wait(ev);
            }
        }
    }

    void task2() {
        while(true) {
            wait(2, SC_SEC);
        }
    }
}
```

```
GNU nano 8.7.1 CMakeLists.txt
cmake_minimum_required(VERSION 3.10)

project(BinarySemaphore)

find_path(SYSTEMC_INCLUDE_DIR systemc.h)
find_library(SYSTEMC_LIBRARY NAMES systemc)

message(STATUS "SystemC include: ${SYSTEMC_INCLUDE_DIR}")
message(STATUS "SystemC library: ${SYSTEMC_LIBRARY}")

add_executable(binary_semaphore binary_semaphore.cpp)

target_include_directories(binary_semaphore PRIVATE ${SYSTEMC_INCLUDE_DIR})
target_link_libraries(binary_semaphore ${SYSTEMC_LIBRARY})
```

```
anish@DESKTOP-BUFCRBD: ~  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ cmake ..  
-- The C compiler identification is GNU 15.2.0  
-- The CXX compiler identification is GNU 15.2.0  
-- Detecting C compiler ABI info  
-- Detecting C compiler ABI info - done  
-- Check for working C compiler: /usr/bin/cc - skipped  
-- Detecting C compile features  
-- Detecting C compile features - done  
-- Detecting CXX compiler ABI info  
-- Detecting CXX compiler ABI info - done  
-- Check for working CXX compiler: /usr/bin/c++ - skipped  
-- Detecting CXX compile features  
-- Detecting CXX compile features - done  
-- SystemC include: /usr/include  
-- SystemC library: /usr/lib/x86_64-linux-gnu/libsystemc.so  
-- Configuring done (3.5s)  
-- Generating done (0.0s)  
-- Build files have been written to: /home/anish/systemc_test/build  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ make  
[ 50%] Building CXX object CMakeFiles/binary_semaphore.dir/binary_semaphore.cpp.o  
[100%] Linking CXX executable binary_semaphore  
[100%] Built target binary_semaphore  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ make  
[100%] Built target binary_semaphore  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ ./binary_semaphore  
  
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46  
Copyright (c) 1996-2025 by all Contributors,  
ALL RIGHTS RESERVED  
Task1 Entered Critical Section at 1 s  
Task1 Entered Critical Section at 4 s  
Task1 Entered Critical Section at 7 s  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ |
```